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Github

Hyeongjin Kim

Education

2017–2021 **Williams College**, *Williamstown, MA.*
Bachelor of Arts in Physics; Cumulative GPA: 3.97/4.00
Dean's List: Fall 2017, Spring-Fall 2018, Spring-Fall 2019, Spring 2020

2010–2016 **Kristin School**, *Auckland, New Zealand.*
International Baccalaureate Diploma; IB Score 44/45
Class of 2016 Valedictorian

Research Experience

June 2019 – Present **Strauch Lab: Optimization and Noise Analysis of Quantum Logic Gates**.
Williams College Physics Department
Honors Thesis Student, Research Assistant
Advisor: Professor Frederick W. Strauch

- Investigated the three-level analyses of the transmon qubit systems for various pulses and developed analytical models of gates with a paper currently in progress.
- Investigated and numerically simulated/optimized a particular fast and high-fidelity gate (denoted as $\sqrt{\varphi\text{–SWAP}}$) for a system of two parametrically coupled, asymmetric transmons.
- Researched the first-principles analysis of transmon circuits and the Derivative Removal by Adiabatic Gate formalism to investigate the optimal parameters of our gate pulses.

Feb 2018 – Present **Jensen Lab: Dynamics of adhesive wetout**.
Williams College Physics Department
Research Assistant
Advisor: Assistant Professor Katharine E. Jensen

- Investigated the mechanics of adhesive contacts of rigid glass spheres with silicone gel surfaces of varying Young's modulus via an co-independently developed two-dimensional image analysis MATLAB program.
- Synthesized silicone gels of varying stiffness, prepared probes with rigid glass spheres, and used an optical microscope with a high-speed camera to investigate adhesive dynamics.
- Assisted in the construction of a Michelson interferometric microscope for the three-dimensional imaging of high-speed interface dynamics.

Publications

2020 Hyeongjin Kim, Frederick W. Strauch, "Fast and high-fidelity quantum logic gates via circuit synthesis and optimal control." In Progress.

Teaching Assistantships

Fall 2020 **Quantum Physics**, *Williams College Physics Department.*
Will host sessions and grade assignments on wavefunctions, solving Schrödinger's equation, analyzing various 1D/2D quantum systems, and analytically modelling the H atom.

Spring 2020 **Algorithm Design and Analysis**, *Williams College Computer Science Department.*
Hosted homework sessions and grade assignments on efficient, provably correct algorithms dealing with divide-and-conquer, dynamic programming, NP-hardness, and randomized approximation methods.

Spring 2019 **Mathematical Methods for Scientists**, *Williams College Physics Department.*
Hosted homework sessions and graded assignments on ordinary and partial differential equations, special functions (including Bessel and Legendre), and numerical methods and simulations in Mathematica.

Presentations

Jul & Aug 2019 Williams College Summer Science Tea Talk and Poster, "Fast and High-Fidelity Quantum Logic Gates for Parametrically Coupled Transmons." Click here for pdf.

Jul & Aug 2018 Williams College Summer Science Tea Talk and Poster, "Dynamics of adhesive wetout and detachment."

Jul 2018 UMass Amherst Soft Matter Day, "Dynamics of adhesive wetout and detachment."

Awards

Aug 2020 Phi Beta Kappa Membership: Top 5% of the class, ranked by cumulative GPA.

2018 & 2019 & 2020 Williams College Summer Science Research Fellowships: Funded summer research experience supervised by Katherine Jensen (2018) and Frederick Strauch (2019 & 2020).

Relevant Coursework

Physics Classical Mechanics; Controlling Quanta; Quantum Physics; Electromagnetic Theory; Vibrations, Waves and Optics; Electricity and Magnetism; Mathematical Methods for Scientists; Modern Physics.

Mathematics Measure Theory and Hilbert Spaces; Representation Theory; Real Analysis; Probability; Applied Partial Differential Equations; Abstract Algebra; Linear Algebra; Discrete Mathematics; Multivariable Calculus.

Computer Science Algorithm Design and Analysis; Data Structures and Advanced Programming; Introduction to Computer Science.

Community Engagement

Jun 2020 – Present **Co-Chair**, *Williamstown, MA.*
Williams College Society of Physics Students
- Organizing and hosting departmental events for physics students, i.e. a LaTeXworkshop.
- Mentoring and helping underclassmen in physics.

2018–2019 **Finance Committe Member**, *Williamstown, MA.*
Williams College Council
- Analyzed budgets and constructed optimal funding and allocation strategies of the College Council budget for all student organizations requesting for funding.

Skills

Languages Python, Java, LaTeX

Tech Mathematica, MATLAB